

Experiment No-2

Mathematical Functions of MATLAB

AIM: The lab aimed to explore the diverse set of mathematical functions available in MATLAB and understand their applications. Through hands-on exercises, various functions, including trigonometric, logarithmic, and basic arithmetic operations, were investigated. The findings contribute to a deeper understanding of MATLAB's capabilities for mathematical computations.

Software Required: Matlab software

Theory:

Mathematical functions

MATLAB offers many predefined mathematical functions for technical computing which contains a large set of mathematical functions.

Typing `help elfun` and `help specfun` calls up full lists of *elementary* and *special* functions respectively.

There is a long list of mathematical functions that are *built* into MATLAB. These functions are called *built-ins*. Many standard mathematical functions, such as $\sin(x)$, $\cos(x)$, $\tan(x)$, e^x , $\ln(x)$, are evaluated by the functions `sin`, `cos`, `tan`, `exp`, and `log` respectively in MATLAB.

Table 2.1 lists some commonly used functions, where variables x and y can be numbers, vectors, or matrices.

Table 2.1: Elementary functions

<code>cos(x)</code>	Cosine	<code>abs(x)</code>	Absolute value
<code>sin(x)</code>	Sine	<code>sign(x)</code>	Signum function
<code>tan(x)</code>	Tangent	<code>max(x)</code>	Maximum value
<code>acos(x)</code>	Arc cosine	<code>min(x)</code>	Minimum value
<code>asin(x)</code>	Arc sine	<code>ceil(x)</code>	Round towards $+\infty$
<code>atan(x)</code>	Arc tangent	<code>floor(x)</code>	Round towards $-\infty$
<code>exp(x)</code>	Exponential	<code>round(x)</code>	Round to nearest integer
<code>sqrt(x)</code>	Square root	<code>rem(x)</code>	Remainder after division
<code>log(x)</code>	Natural logarithm	<code>angle(x)</code>	Phase angle
<code>log10(x)</code>	Common logarithm	<code>conj(x)</code>	Complex conjugate

A list of the most common values is given in Table 2.2.

Table 2.2: Predefined constant values

pi	The π number, $\pi = 3.14159\dots$
i,j	The imaginary unit i , $\sqrt{-1}$
Inf	The infinity, ∞
NaN	Not a number

Examples

We illustrate here some typical examples which related to the elementary functions previously defined.

As a first example, the value of the expression $y = e^{-a} \sin(x) + 10\sqrt{y}$, for $a = 5$, $x = 2$, and $y = 8$ is computed by

```
>> a = 5; x = 2; y = 8;
>> y = exp(-a)*sin(x)+10*sqrt(y)
y =
28.2904
```

The subsequent examples are

```
>> log(142)
ans =
4.9558
```

```
>> log10(142)
ans = 2.1523
```

Note the difference between the natural logarithm $\log(x)$ and the decimal logarithm (base 10) $\log_{10}(x)$.

To calculate $\sin(\pi/4)$ and e^{10} , we enter the following commands in MATLAB,

```
>> sin(pi/4)
ans =
0.7071
```

```
>> exp(10)
ans      =
2.2026e+004
```

Creating simple plots

The basic MATLAB graphing procedure, for example in 2D, is to take a vector of x -coordinates, $\mathbf{x} = (x_1, \dots, x_N)$, and a vector of y -coordinates, $\mathbf{y} = (y_1, \dots, y_N)$, locate the points (x_i, y_i) , with $i = 1, 2, \dots, n$ and then join them by straight lines. You need to prepare x and y in an identical array form; namely, x and y are both row arrays or column arrays of the *same* length.

The MATLAB command to plot a graph is `plot(x,y)`. The vectors $\mathbf{x} = (1, 2, 3, 4, 5, 6)$ and $\mathbf{y} = (3, -1, 2, 4, 5, 1)$ produce the picture shown in Figure 2.1.

```
>> x = [1 2 3 4 5 6];
>> y = [3 -1 2 4 5 1];
>> plot(x,y)
```

NOTE: The `plot` functions has different forms depending on the input arguments. If y is a vector `plot(y)` produces a piecewise linear graph of the elements of y versus the index of the elements of y . If we specify two vectors, as mentioned above, `plot(x,y)` produces a graph of y versus x .

For example, to plot the function $\sin(x)$ on the interval $[0, 2\pi]$, we first create a vector of x values ranging from 0 to 2π , then compute the *sine* of these values, and finally plot the result:

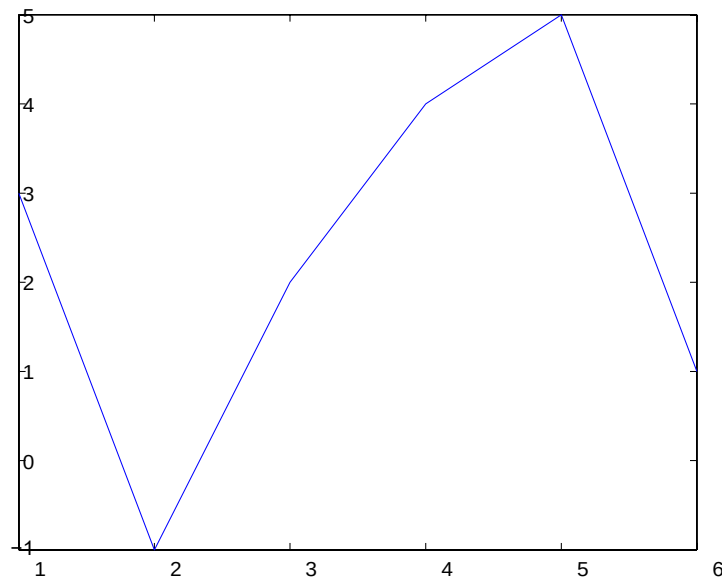


Figure 2.1: Plot for the vectors x and y

```
>> x = 0:pi/100:2*pi;
>> y = sin(x);
>> plot(x,y)
```

Adding titles, axis labels, and annotations.

Now label the axes and add a title. The character `\pi` creates the symbol π . An example of 2D plot is shown in Figure 2.2.

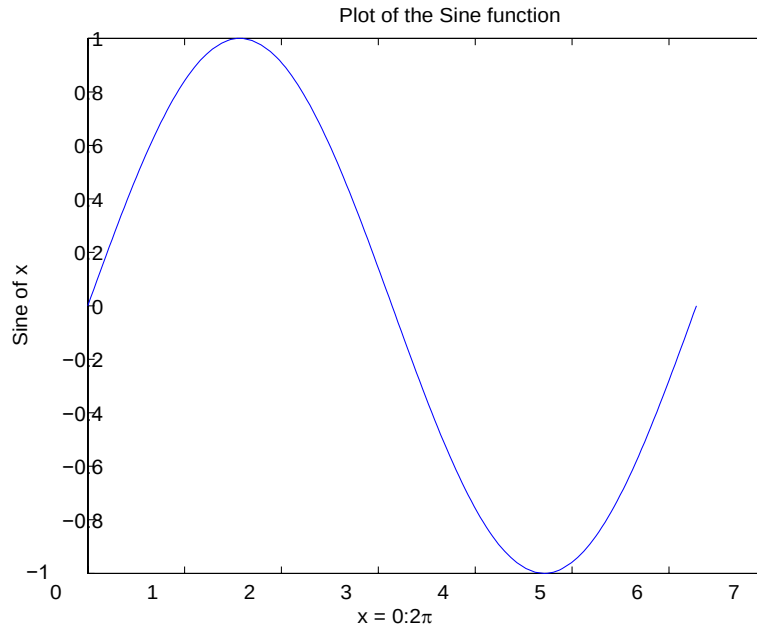


Figure 2.2: Plot of the Sine function

```
>> xlabel('x = 0:2\pi')
>> ylabel('Sine of x')
>> title('Plot of the Sine function')
```

The color of a single curve is, by default, blue, but other colors are possible. The desired color is indicated by a third argument. For example, red is selected by `plot(x,y,'r')`. Note the single quotes, ' ', around r.

Multiple data sets in one plot

Multiple (x, y) pairs arguments create *multiple* graphs with a single call to `plot`. For example, these statements plot three related functions of x : $y_1 = 2\cos(x)$, $y_2 = \cos(x)$, and $y_3 = 0.5 * \cos(x)$, in the interval $0 \leq x \leq 2\pi$.

```
>> x = 0:pi/100:2*pi;
>> y1 = 2*cos(x);
>> y2 = cos(x);
>> y3 = 0.5*cos(x);
>> plot(x,y1,'--',x,y2,'-',x,y3,':')
>> xlabel('0 \leq x \leq 2\pi')
>> ylabel('Cosine functions')
```

```
>> legend('2*cos(x)', 'cos(x)', '0.5*cos(x)')
>> title('Typical example of multiple plots')
>> axis([0 2*pi -3 3])
```

The result of multiple data sets in one graph plot is shown in Figure 2.3.

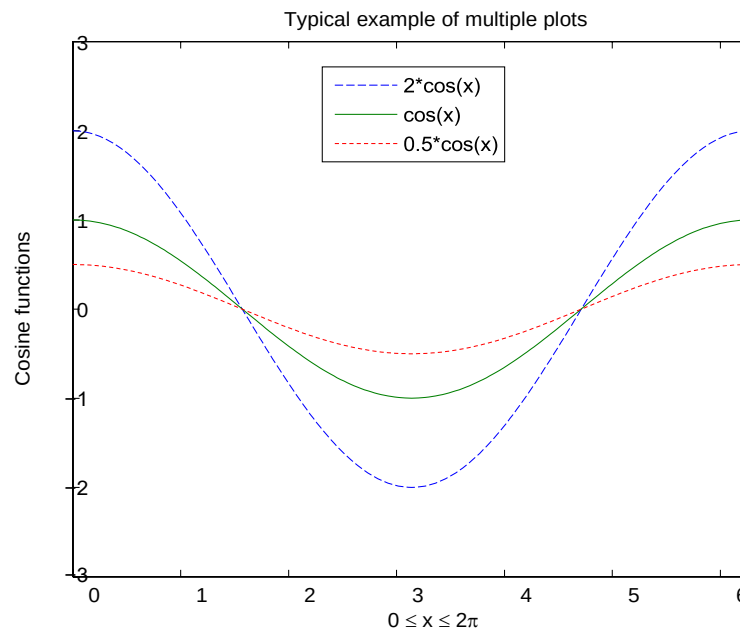


Figure 2.3: Typical example of multiple plots

By default, MATLAB uses *line style* and *color* to distinguish the data sets plotted in the graph. However, you can change the appearance of these graphic components or add annotations to the graph to help explain your data for presentation.

Specifying line styles and colors

It is possible to specify *line styles*, *colors*, and *markers* (e.g., circles, plus signs, ...) using the plot command:

```
plot(x,y,'style_color_marker')
```

where *style_color_marker* is a *triplet* of values from Table 2.3.

Table 2.3: Attributes for plot

SYMBOL	COLOR	SYMBOL	LINE STYLE	SYMBOL	MARKER
k	Black	—	Solid	+	Plus sign
r	Red	--	Dashed	•	Circle
b	Blue	:	Dotted	*	Asterisk
g	Green	-.	Dash-dot	.	Point
c	Cyan	none	No line	×	Cross
m	Magenta			s	Square
y	Yellow			d	Diamond